Bank Marketing Campaign

Understanding Customer Subscription Patterns

Full Exploratory Data Analysis (EDA) Report

What we did, questions answered, and findings

This report explains the whole analysis: dataset understanding, cleaning decisions, the 8 questions, univariate and bivariate work, answers to each question, and the final recommendation.

1. Project goal

A Portuguese bank ran phone marketing campaigns to sell term deposits. We analysed customer and campaign data to learn who is more likely to subscribe after a marketing call.

Big question

Which customer and campaign factors are linked to a term-deposit subscription?

2. Dataset understanding

Source: UCI / Kaggle Bank Marketing dataset (file: data/bank.csv). One row = one customer contact during the campaign.

- Rows: 11,162
- Columns: 17
- Target column: deposit (yes / no)
- Overall subscription rate: about 47.4% (5,289 yes / 5,873 no)

Important columns

- age, job, marital, education = customer profile
- balance, housing, loan, default = financial situation
- contact, day, month, duration, campaign = this campaign
- pdays, previous, poutcome = previous campaign history
- deposit = whether the customer subscribed (TARGET)

3. Questions we asked (8)

Questions were chosen so the dataset can actually answer them.

Univariate (one column)

- Q1. What is the overall subscription rate (deposit)?
- Q2. How is customer age distributed?
- Q3. Which jobs appear most often in the campaign?

Bivariate (two columns)

- Q4. How does call duration relate to subscription?
- Q5. Do customers with a housing loan subscribe less often?
- Q6. Which contact channel has the highest subscription rate?

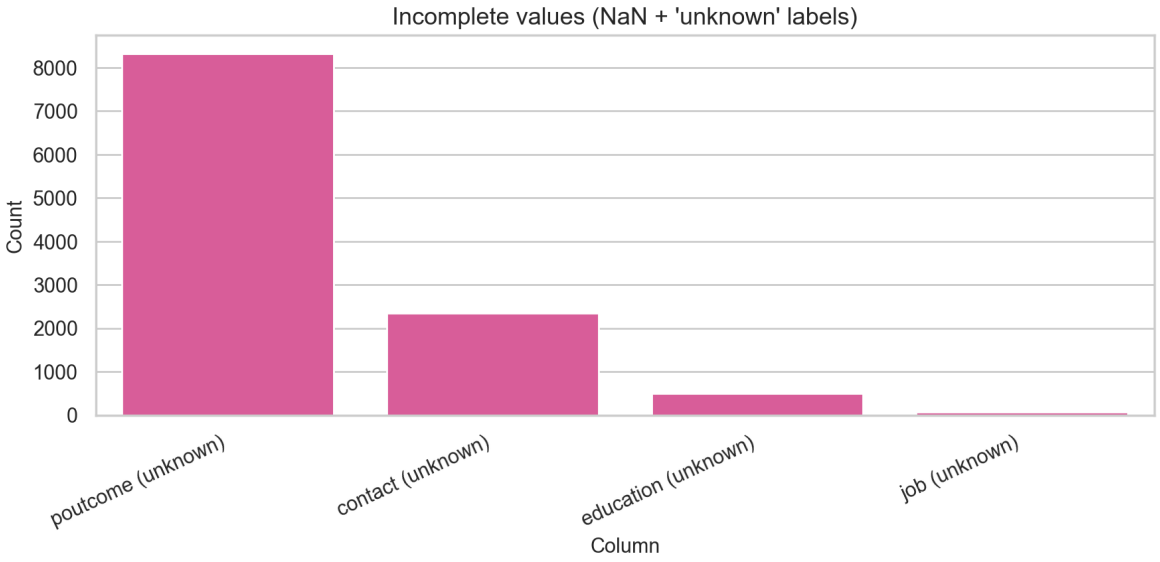
Group comparison

- Q7. How does subscription rate compare across education groups?
- Q8. How do job and marital status groups differ in subscription rate?

4. Data cleaning

4.1 Missing values

True NaN missing values: 0. Some fields use the label 'unknown' (job, education, contact, poutcome). We kept 'unknown' as its own category. Decision: KEEP. Dropping unknown would remove thousands of valid contacts (especially poutcome) and bias the campaign view.



4.2 Repeated rows

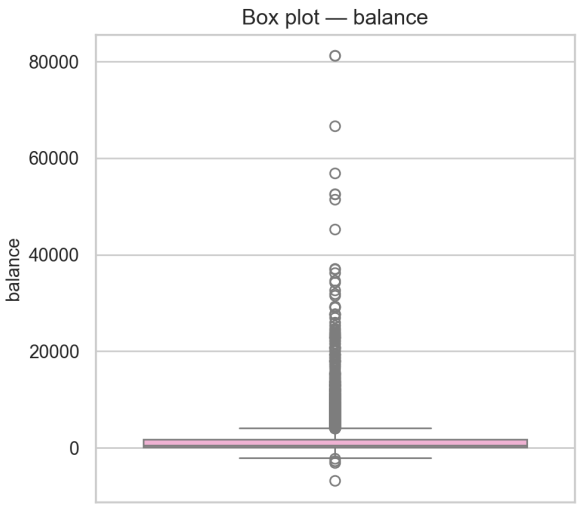
Duplicate rows counted with df.duplicated().sum() = 0. Decision: nothing to remove. Proof: zero exact repeated rows.

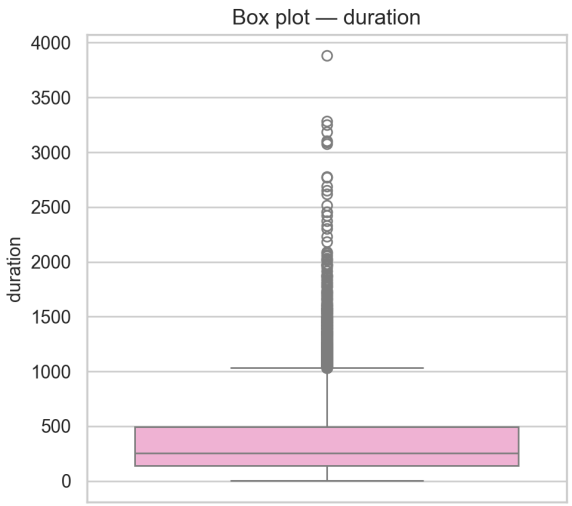
4.3 Data types

Numeric fields (age, balance, duration, etc.) were already integers. Month text was converted to an ordered category (jan to dec). Yes/no columns got 0/1 helper flags for correlation analysis. There is no full calendar date column in this dataset.

4.4 Outliers (IQR on balance and duration)

Box plots and the IQR rule flagged many high balances and long call durations. Decision: KEEP outliers. High balances look like real wealthy/indebted customers; long durations look like real long calls. Removing them would hide important high-conversion patterns.

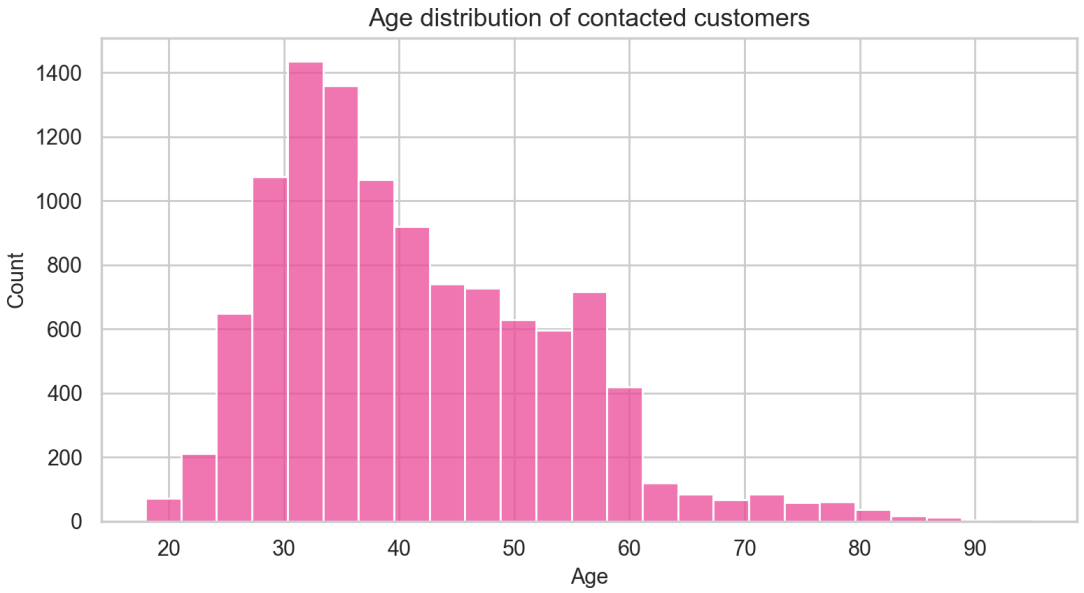

Box plot — balance
balance
80000
60000
40000
20000
0

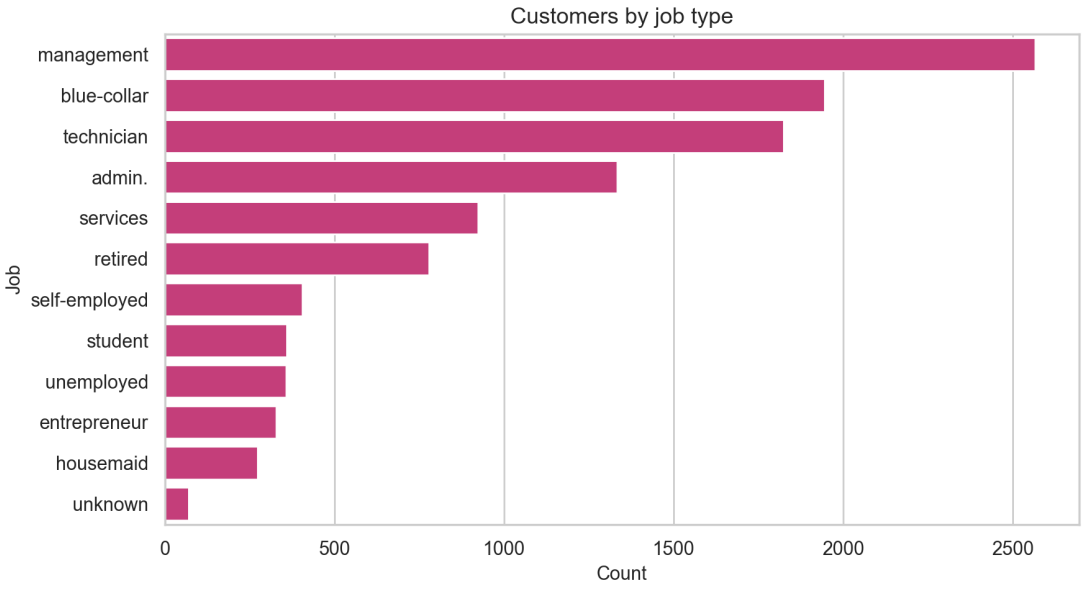

Box plot — duration
duration
4000
3500
3000
2500
2000
1500
1000
500
0

5. Univariate analysis

We plotted histograms for numeric columns and count charts for categorical columns.

- Age peaks in the 30s-40s (mean age about 41).
- Balance and duration are right-skewed.
- Most customers have few campaign contacts; many have pdays = -1.
- Top jobs: management, blue-collar, technician.
- Most customers are married with secondary education.
- Cellular is the main contact channel; May has the most contacts.

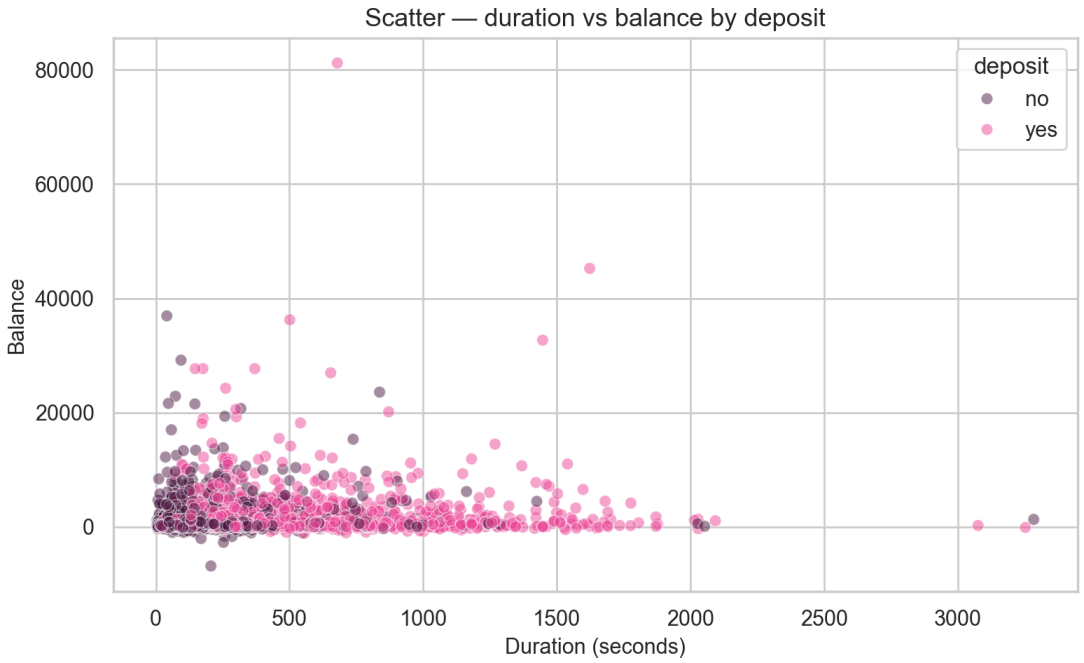


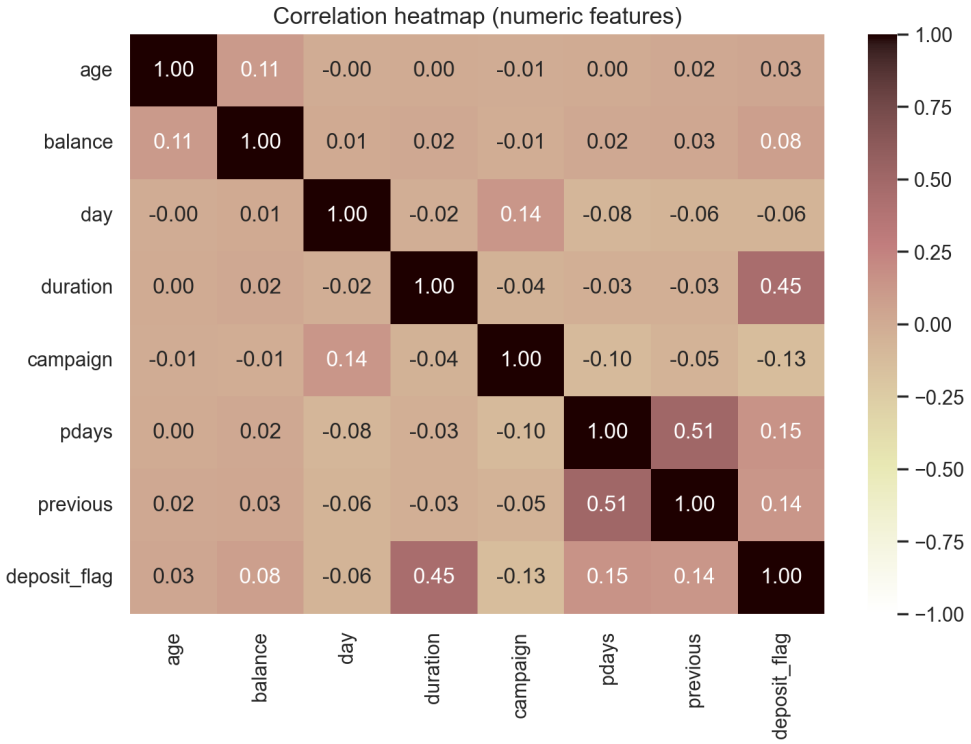


6. Bivariate analysis

We used scatter plots, a correlation heatmap, and category comparisons (box plots / bar charts).

- Strongest numeric link with subscription: call duration.
- Longer calls are denser among deposit = yes.
- Balance alone does not cleanly separate yes/no groups.
- Subscribers have a clearly higher duration median than non-subscribers.

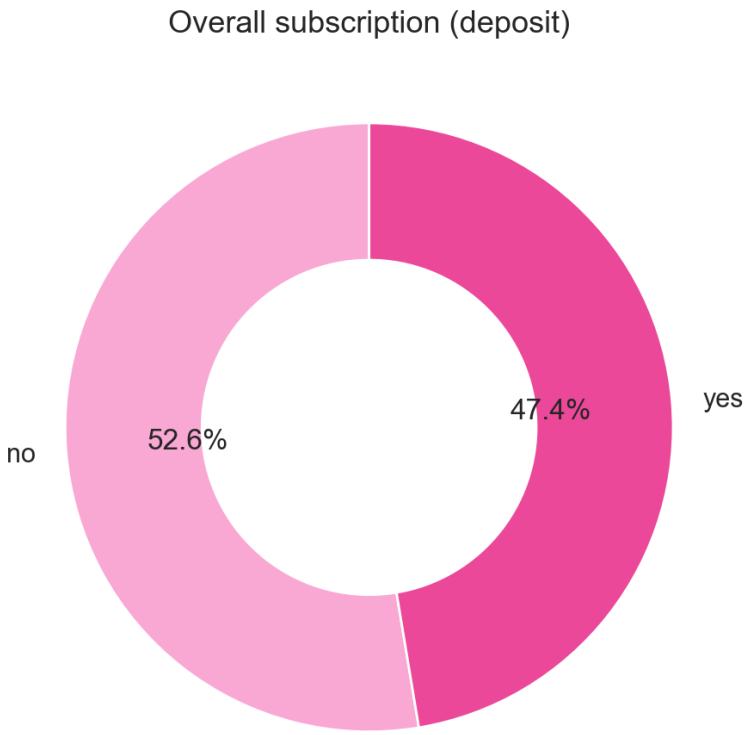




7. Answers to each question

Q1. Overall subscription rate?

About 47.4% of contacted customers subscribed. The campaign outcome is fairly balanced (not a rare-event problem).



Q2. Age distribution?

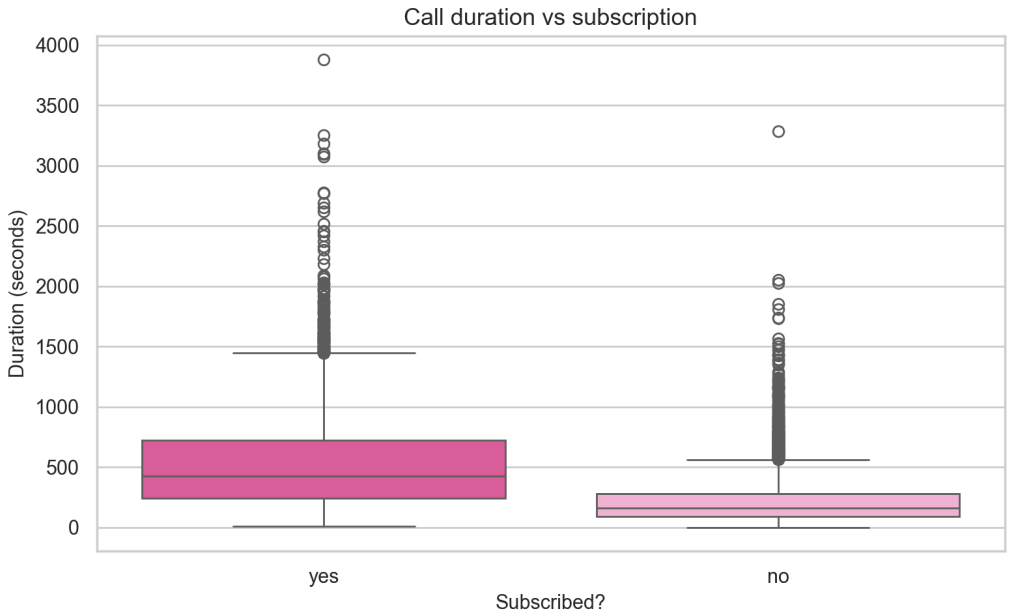
Ages centre around the late 30s / early 40s. Most contacts are working-age adults, with fewer teens and seniors.

Q3. Which jobs appear most?

Management, blue-collar, and technician are the three most contacted job groups.

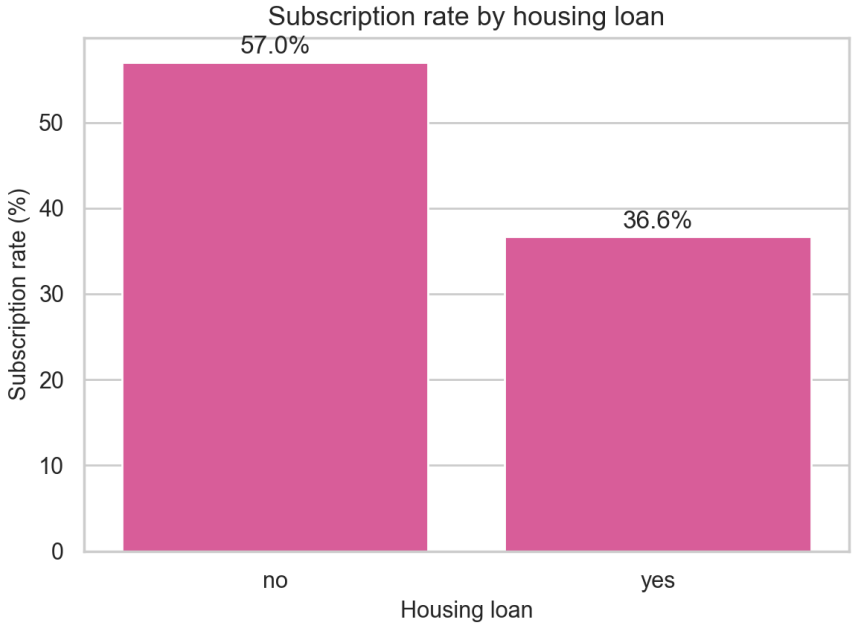
Q4. Duration vs subscription?

Subscribers have a much higher median call duration (about 426 seconds vs 163 seconds). Longer conversations strongly track with a 'yes'. Note: duration is known after the call, so it is a strong descriptive signal but should be used carefully for prediction.


Call duration vs subscription
Duration (seconds)
4000
3500
3000
2500
2000
1500
1000
500
0
yes
no
Subscribed?

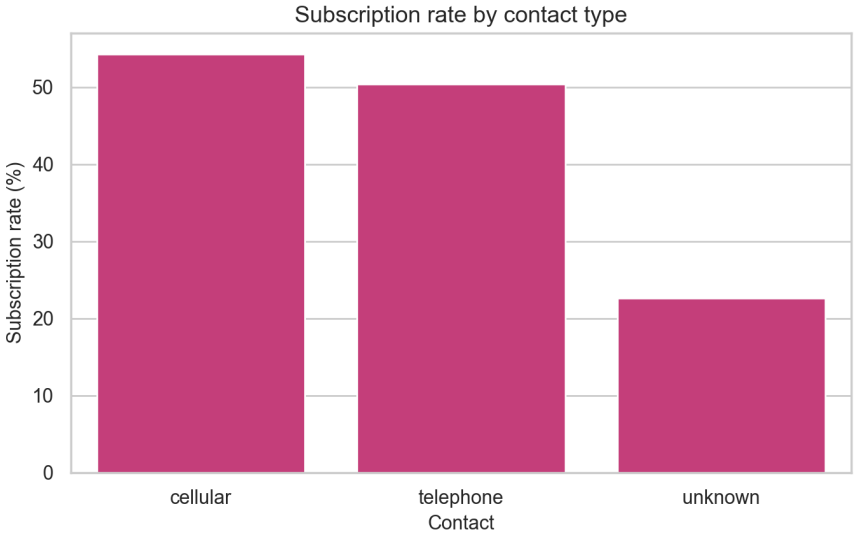
Q5. Housing loan effect?

Customers without a housing loan subscribe at a clearly higher rate (about 57%) than those with a housing loan (about 37%).



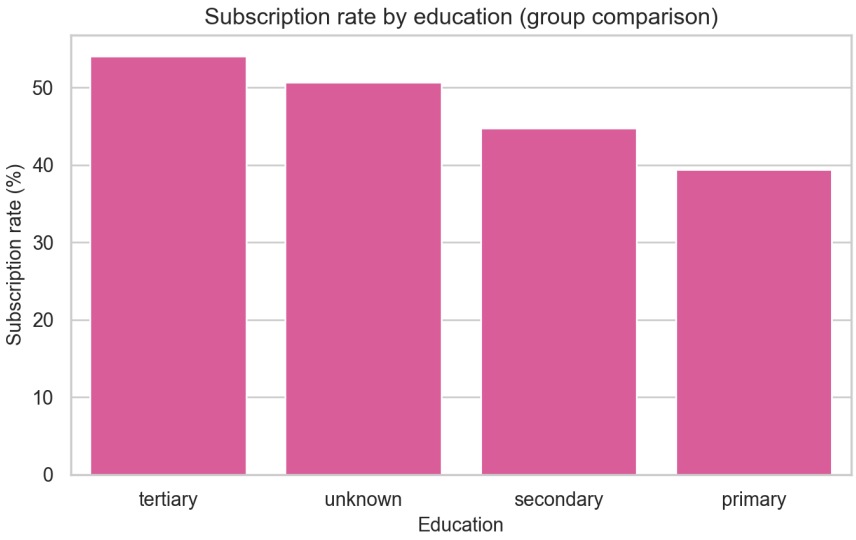
Q6. Best contact channel?

Cellular and telephone outperform 'unknown' contact. Known phone channels convert better.



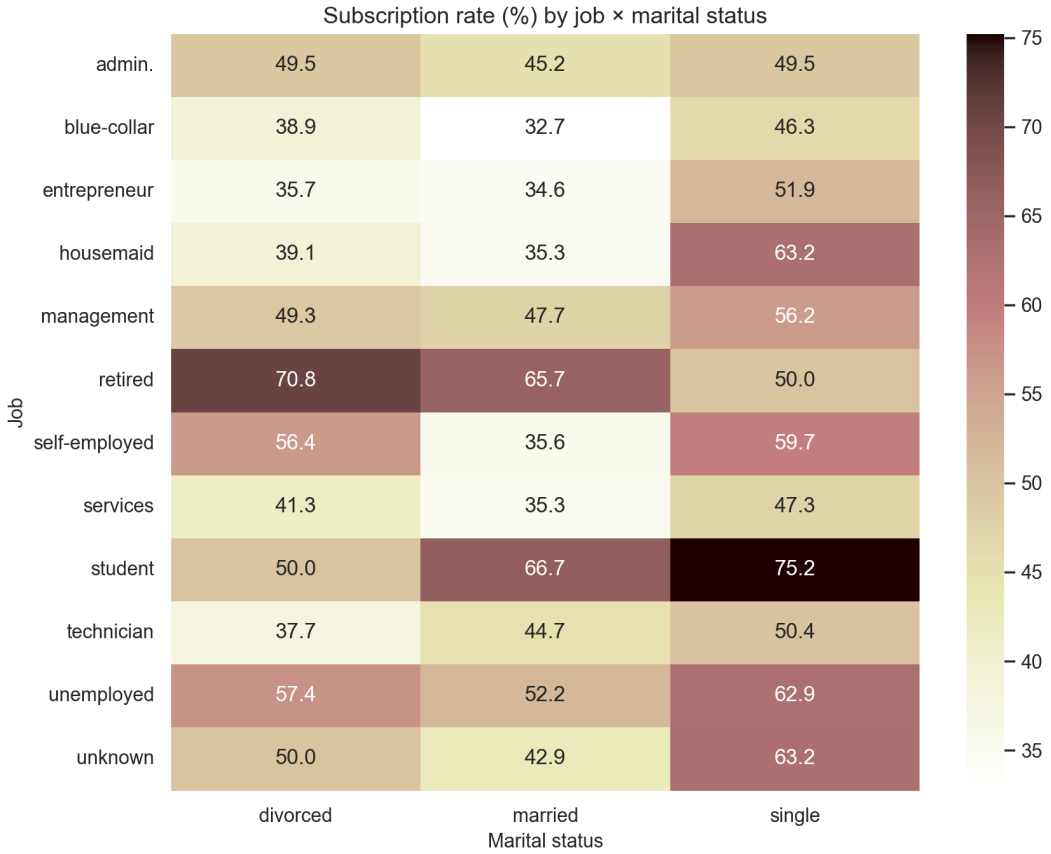
Q7. Education groups?

Tertiary education shows the highest subscription rate among education groups; primary is lower.



Q8. Job and marital groups?

Subscription rates vary widely by segment. Students and retirees often convert higher, while some blue-collar/married cells are lower. Targeting should be segment-specific.



8. Conclusion

Three main findings

- Call duration is the strongest numeric signal linked to subscription.
- Customers without a housing loan subscribe more often than those with one.
- Education, job, and contact channel create clear conversion differences.

One actionable suggestion

Prioritise cellular outreach to customers without housing loans, and train agents to hold high-quality conversations (engagement/duration) rather than simply increasing the number of campaign contacts.

9. Project deliverables

- Notebook: Bank_Marketing_EDA.ipynb (full Steps 1-7 with explanations)
- Dashboard: app.py (Streamlit filters, KPIs, Plotly charts, findings)
- Data: data/bank.csv
- Slide charts: slides_assets/
- This PDF: Bank_Marketing_Analysis_Report.pdf

End of report. Open the notebook for full code, and the Streamlit app for interactive filters.